

Q. Java generates hash code for every object. Find out what is the hash code in java and the purpose of generating the hash code.

Ans

A hash code is a 32-bit integer value generated by the `hashCode()` method of the `Object` class. Every Java object has a hash code. It is mainly used to identify objects efficiently and to support fast searching in hash-based collections.

The default implementation of `hashCode()` generates a value based on the object's memory-related identity. However, classes such as `String`, `Integer`, and user-defined classes can override the `hashCode()` method to generate a hash code based on the object's contents.

Ex: class Demo of

```
public static void main(String[] args) {  
    String s = "java";  
    System.out.println(s.hashCode());  
}
```


Purpose of generating Hash Code:

1. fast searching: enables quick searching and retrieval
2. ~~Used~~ of objects
2. Used in hash-based collections: Used by HashMap, HashSet, Hashtable, and LinkedHashMap.
3. Efficient storage: helps place objects into the correct bucket, reducing search time.
4. Improves performance: Makes insertion, deletion, and lookup operations faster.
5. Works with equals(): If two objects are equal according to equals(), they must have the same hash code.